Response to Referee R1

We sincerely thank you for your time, effort, and valuable feedback. Your detailed comments and constructive suggestions have significantly improved the paper. We have carefully considered each point and prepared a thorough, point-by-point response. Below, we respond to each comment individually; for ease of reading, the original comments are italicized and displayed in red. Before addressing each comment, we begin with a brief summary of the major revisions.

- **Extended sample.** Following R1's and R6's suggestions, we extended our panel data coverage from seven years (1930–1936) to fifteen years (1926–1940). This extension serves two primary purposes:

  1. Examining additional pre-treatment years (1926–1932 rather than just 1930–1932) enables a more rigorous evaluation of the parallel trends assumption underlying our difference-in-differences design.
  2. The recession of 1937–1938 provides a natural placebo test: if our gold-clause exposure measure captured something other than the mechanism we propose—such as general sensitivity to economic downturns—we would expect it to affect investment during this later period of distress as well.

  Figure R.I (Figure 3 in the revised manuscript) plots the year-by-year interaction coefficients from our main investment regression over the extended sample. The figure confirms the absence of pre-trends before 1933, sharp negative effects during the 1933–1934 litigation period, and no effect during the 1937–1938 recession.

- **Data reconstruction.** While extending the sample, we identified that some scans purchased from Mergent were of poor quality, leading to firm omissions and data entry errors. To address these concerns, we undertook a complete data reconstruction:

  1. We scanned, in high resolution, the Moody's Industrial Manuals for all sixteen years, ensuring complete legibility of all pages.

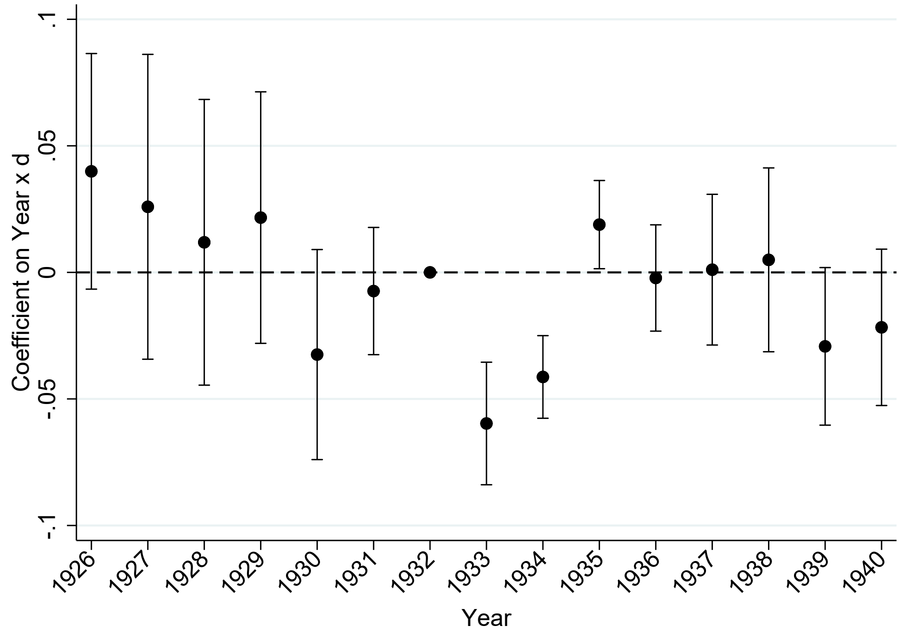

Figure R.I. Gold clause effect on investment by year

Notes. This figure plots point estimates and 95% confidence bands from a regression of net investment on Q and $\tilde{d}$ interacted with year indicators, including firm and year fixed effects.

2. We recollected the entire dataset using a double-blind procedure in which two independent research assistants collected data separately, with discrepancies resolved by a third party.

3. We implemented extensive quality control checks to identify and correct any remaining issues.

This reconstruction process required nearly two years to complete but was essential for ensuring the reliability of our findings.

- **Gold clause exposure measure.** Following R1's suggestion, we replaced total debt with long-term liabilities in the denominator of our gold exposure measure, $d_{j,t}$, now defined as the ratio of gold-clause bonds to long-term liabilities (corporate bonds, preferred shares, and bank loans) for firm j at time t:

$$d_{j,t} = \frac{\text{Gold-denominated long-term liabilities of firm } j \text{ in year } t}{\text{Long-term liabilities of firm } j \text{ in year } t}.$$

- **Addressing endogeneity concerns.** Following R6's suggestion, we fix each firm's gold exposure at its 1930 value for all subsequent years to address potential endogeneity from strategic adjustments following Britain's departure from the gold standard in September 1931. Formally:

$$\tilde{d}_{j,t} = \begin{cases} d_{j,t}, & \text{if } t \leq 1930 \\ d_{j,1930}, & \text{if } t > 1930 \end{cases}.$$

  This measure, $\tilde{d}$, is our main explanatory variable throughout the revised analysis.

- **Empirical design.** Following R1's, R2's, and R6's suggestions, we now use the full 1926–1940 panel and estimate the effects of gold clause exposure and its reversal in a single regression.

- **Debt overhang framing.** Following R1's and R2's suggestions, we have substantially revised the paper's conceptual framing. We no longer use the term "leverage risk" in the paper title or text and instead interpret the events of 1933–1934 through the lens of dynamic debt overhang theory (Hennessy, 2004). This framing more precisely captures the economic mechanism linking gold clause exposure to investment decisions.

- **Firm-level uncertainty literature.** Following R2's suggestion, we have repositioned the paper within the firm-level uncertainty literature (Bloom, 2009; Baker et al., 2016; Hassan et al., 2019). The uncertainty surrounding the gold clause represents a policy shock of the type studied extensively in this growing literature. This reframing highlights that our gold clause exposure measure, $\tilde{d}_{j,t}$, captures firm-level variation in exposure to policy uncertainty—derived from balance sheet data rather than earnings call transcripts as in (Hassan et al., 2019). We are grateful to R2 for this suggestion, as it clarifies how our historical setting illustrates the real effects of policy uncertainty during a pivotal episode.

Major Comment 1.

The authors conduct a placebo test which includes preferred stock and "bond amount outstanding without gold clauses in total liabilities". My impression though is that virtually all corporate bonds of any importance/size at this time included gold clauses. What percent of corporate bonds (by # and $) in this sample are 100% confirmed to not have gold clauses? I would think that number, especially in $ terms, is tiny.

Thank you for raising this important point. Your impression is correct: in 1932, about 97% of the corporate bonds in our sample by count (99% by outstanding par value) had the gold clause. Put differently, only 3% of bonds by count and 1% by dollar value were confirmed to lack gold clauses—numbers that are indeed tiny, as you suspected. In our original submission, we had this information in a footnote, but we agree that such an important piece of information should be featured more prominently in the paper.

To clarify this point, we now mention it on the first page of the introduction: "When the United States effectively abandoned the gold standard in April 1933, the dollar quickly depreciated against gold, and the existence of these clauses posed a serious threat to most corporate bond issuers: around 97% of outstanding corporate bonds were subject to the gold clause." We hope that explicitly stating the exact number early in the introduction leaves no room for ambiguity and helps frame the economic significance of the gold clause from the outset.

Given this near-universal prevalence of gold clauses in corporate bonds, much of the variation in our measure of gold exposure comes from the composition of firms' liabilities—specifically, the relative share of corporate bonds versus preferred stock. Since preferred shares uniformly lacked gold clauses while nearly all corporate bonds contained them, firms financing with more bonds relative to preferred stock have higher gold clause exposure. As we discuss in our response to Comment 2.d, we have adopted your suggestion to define d as the share of gold clause bonds in long-term debt (bonds plus preferred shares plus bank debt), which directly captures this source of variation. We clarify this when defining our exposure measure in equation (1) and in Section 4.

We have added details on the total number of bonds in our sample and on the number of bonds containing gold clauses. These results are reported in Table 2 and discussed in Section 4 of the revised manuscript. We also emphasize this point when defining gold clause exposure in equation (1), noting that nearly all bonds issued before the leverage uncertainty period included a gold clause.

On a related note, what percent of all gold clause liabilities, as measured in this paper, come from corporate bonds vs any other liabilities (# and $ terms)? Again, my impression from looking through the Moody's Manuals is that virtually all the information on inclusion of gold clauses would be from long-term corporate bonds.

Your impression is again correct. In our measure d, 100% of gold-denominated liabilities (the numerator) come from long-term corporate bonds having a gold clause. Preferred shares uniformly lacked gold clauses, and while it is technically possible that some bank loans also contained gold clauses, we do not have information on their presence. In any case, bank debt accounts for a negligible fraction of the capital structure of most public firms in our sample. As noted above, this is precisely why variation in d reflects the composition of long-term financing—bonds versus preferred shares—rather than variation in whether bonds contain gold clauses. As we now state in Section 3, "[w]hile the clause is present in the vast majority of corporate bonds (97% in our sample), it is entirely absent from preferred shares, another type of long-term obligations commonly used at the time."

If I'm mistaken and there is a lot of other variation it would helpful for the authors to clarify that and demonstrate it in the text. It would also suggest that using the % of long-term corporate bonds with gold clauses, would be a much more natural workhorse empirical design and I would encourage the authors to use that. If, however, as appears to be the case - nearly all the variation in gold clause liabilities is variation in the amount of bonds – that needs to stated and shown explicitly to the reader. To understand the validity of the design it is important to know that the primary source of variation is comparing firms with and without bonds, or with differing amount of bonds relative to their other liabilities, as opposed to firms who happened to include gold clauses or not in their bonds.

We agree that this point was not sufficiently emphasized in the original submission. As you correctly note, the primary source of variation is comparing firms with differing amounts of bonds relative to their other long-term liabilities—primarily preferred shares—rather than comparing firms whose bonds happened to include or exclude gold clauses. In the revised manuscript, we state this explicitly when defining d in equation (1) and again in Section 4. Specifically, in Section 4 we write:

> $d_{j,t}$ measures firm j's exposure to the gold clause at time t as the fraction of its long-term liabilities containing the clause:
>
> $$d_{j,t} = \frac{\text{Gold-denominated long-term liabilities of firm } j \text{ in year } t}{\text{Long-term liabilities of firm } j \text{ in year } t}.$$
>
> We define the long-term liabilities of a firm as the sum of the par value of these three components: corporate bonds, preferred shares, and bank loans. To construct $d_{j,t}$, we collect individual security-level information on the presence of the gold clause. As noted in the previous section, the gold clause was included in most corporate bonds (around 97% of them) and absent from all preferred shares. For bank debt, the manuals do not report exposure to the gold clause. However, at the time of the abrogation of the gold clause bank debt only accounts for 1% of long-term liabilities [...].

As discussed in our response to Comment 2.d, we have also adopted your suggestion to use the share of gold clause bonds in long-term debt as our primary specification, which directly reflects this source of variation. We believe these clarifications help readers understand the nature of our identifying variation and the validity of our empirical design.

Major Comment 2.

My biggest concern is the plausibility of the identifying assumptions within the research design of this paper. The implicit assumption here for these estimated effects to be valid is that over the two year period of 1933 and 1934, the only difference of any importance driving variation in the change in net investment for firms with and without long-term corporate bonds (or more vs. less bonded *debt as a % of total liabilities) is leverage risk from the gold clauses they have. This is despite the fact that firms with gold clauses – aka those with bonded debt – are on average more than twice as large, more than 50% more highly levered, have higher payouts, have more non-current assets, and lower profits. While it is comforting that the authors show that their findings are largely unchanged controlling both linearly and non-parametrically for many of these observed differences, it seems almost certain these firms differ on dimensions not analyzed/observed by the authors as well. Since the period examined is the aftermath of the Great Depression and includes the "first 100 days" period of FDR presidency, the establishment of the SEC, the Glass-Steagall Act, and huge numbers of other regulatory changes, market fluctuations, and re-organizations it is likely one of the more challenging periods in economic history to argue for the absence of concurrent confounds. With that in mind, I think there are a number of dimensions along which the authors could try to improve how they address these concerns:*

We fully agree with your assessment. The implicit assumption in our empirical design is that, after controlling for observables and time-invariant unobservable factors, variation in investment rates is driven by firms' exposure to the gold clause. As you point out, this leaves room for time-varying unobservable confounding factors. And as you also note, the Great Depression saw many concurrent policy changes, making this one of the more challenging periods in economic history to study. We now explicitly acknowledge this challenge in the introduction of the revised manuscript.

Despite these methodological difficulties, we believe this episode remains important to study given the Great Depression's profound influence on macroeconomics and finance research. We are grateful for the concrete suggestions you provided to strengthen our identification strategy. As detailed below, we have implemented each of them, and we believe these changes have substantially improved the paper.

a. *Controls for observables: Since these firms differ in terms of their 1931-1932 payout/fixed capital it would be helpful to control for that interacted with the year linearly and nonparametrically just as is done for the other pretreatment values in Table 6. The following additional variables may not need to be in the main paper, but at a minimum in the appendix it would be helpful to include stock return volatility, Fama-French factor beta loadings, sales/assets,* *and location controls (ex. NYC vs. not). It would also be useful to see the full distribution of 1931-1932 variables for d>0, d=0, d<0.44, and d>0.44, even if it is just in an appendix.*

In the revised manuscript, we add payout (now normalized by the book value of common stock) to the set of linear controls interacted with year (Column 2 of Table 6) and also include it nonparametrically, as we did the other controls (Column 7 of Table 6). Following your suggestion, we additionally control for the mean, volatility, and Fama–French three-factor betas, with results reported in Internet Appendix Table IA.17. As discussed in Section 5.5, our results remain robust to these additional controls. The Moody's Manuals are inconsistent in their reporting of sales: sometimes sales are reported as gross revenues, and sometimes as net revenues. This inconsistency is visible both across firms and within the same firm over time, making it impossible to construct a reliable sales measure. For this reason, we were unable to include sales as a control. Finally, Internet Appendix Tables IA.4–IA.7 report the full distribution of firm characteristics in our extended sample for 1926–1932, 1933–1934, and 1935–1940, separately for $d > 0$, $d = 0$, below-median d, and above-median d firms. In sum, adding these controls—whether linearly or nonparametrically—does not materially alter our estimates, reinforcing the conclusion that gold clause exposure, rather than correlated firm characteristics, drove the investment decline during the leverage uncertainty period.

b. *Difference-in-Differences (DiD) Design: One of the primary arguments the authors put forth for why selection on unobservables is unlikely to be an issue is the evidence from their DiD design. That being said, the current implementation doesn't appear particularly compelling and I would encourage the authors to improve this substantially. It would be helpful to see the actual DiD coefficients plotted annually at a minimum all the years the authors state they already have data for (1930-1936). The way things are presented now, in just two-year aggregated increments, it is almost impossible to assess the validity of the DiD assumptions. Even more useful would be for them to extend their dataset to cover a few additional years both before and after the 1933 abrogation and 1935 supreme court decision to show that firms with gold clauses responded with net investment similarly in all years, except the two "treated"* *years of interest in 1933 and 1934. There is obviously no hard rule here, but something like 1925-1940 would at least be somewhat more convincing if DiD coefficients are plotted year-by-year and it is only 1933 and 1934 which are each statistically significantly lower, while all other years there is no differential net investment, relative to the omitted period.*

We sincerely thank you for this comment. We believe your recommendation has significantly improved the paper. Specifically, we made two major updates to our analysis:

1. Following your suggestion, we extended our sample period to 1925–1940 by hand-collecting annual balance sheet and income statement items from the Moody's Industrial Manuals, as described in Section 4. All analyses in the revised manuscript use this extended dataset (1926–1940 to allow for the construction of growth-rate variables). This extension required extensive quality checks and a full re-collection of the earlier sample period to ensure consistency across years, which was the main reason for the delayed resubmission.

2. We now present our baseline results year by year rather than grouping into two-year intervals. In addition, we estimate the effects jointly over the entire sample period in a single regression, rather than running separate regressions for treatment and reversal.

In the revised manuscript, Section 5.2 presents our baseline investment results. These regressions also control for Tobin's Q, motivated by the conceptual framework developed in Section 5.1. Drawing on Hennessy (2004)'s dynamic debt overhang theory, this framework shows that debt overhang drives a wedge between marginal and average Q, making it essential to condition on Q when estimating the effect of gold clause exposure on investment.

Figure 3 of the paper presents the result we believe you have in mind. We reproduced that figure at the beginning of our response, see Figure R.I. The corresponding regression coefficients are reported in Table 3 of the paper. The figure plots the interaction coefficients between $\tilde{d}$ (our measure of gold clause exposure) and year indicators along with 95% confidence intervals, using 1932 as the omitted category. The coefficient pattern shows: (i) no statistically significant relationship between $\tilde{d}$ and investment in the pre-treatment years (1926–1932), supporting the parallel trends assumption; (ii) negative and statistically significant coefficients in 1933 and 1934, the years of gold clause litigation (-0.060 and -0.041, respectively, both significant at the 1% level); and (iii) a return to statistically insignificant coefficients after the Supreme Court's 1935 decision. The absence of effects in 1937–38—another period of economic distress—further validates that gold exposure does not capture general sensitivity to downturns.

We also note that, following a suggestion by R6, $\tilde{d}$ is time-varying before and through 1930 but is fixed at the 1930 value for subsequent years to address potential endogeneity from post-1931 strategic adjustments (following Britain's departure from the gold standard), as discussed in Section 4 of the manuscript. The year-by-year structure also reveals that the investment decline was concentrated in 1933, with a smaller (though still significant) effect in 1934.

Thank you for this suggestion. The fact that 1933 and 1934 stand out as unique relative to all other years from 1926 to 1940 helps address concerns about alternative channels and concurrent confounds. We now adopt this specification throughout the paper and use the full 1926–1940 dataset in all tests.

c. *Secondary market data: Rather than looking at firms with publicly listed common stock on the NYSE (aka what is in CRSP) as a convenience sample, one justification is that there are a number of approaches using secondary market data that could help validate the research design. Monthly estimates of equity volatilities, average returns, volume of trading, bid-ask spreads, Fama-French factor loadings, et al. could all provide evidence of no differential trends in gold clause firms in the months prior to the events of interest, and a reversal that lines of up with what would be expected.*

Focusing on public firms limits our sample but is conceptually essential: our investment–Q regressions in the presence of debt overhang require market-based measures of firm value.

These regressions play a larger role in the revised manuscript and help interpret our results. We clarify this motivation at the beginning of Section 4.

That said, you are right that secondary market equity data can help validate our research design. We address this in two ways. First, as detailed in our response to Comment 2.a, we incorporate stock return characteristics (mean, volatility, and Fama–French betas) as additional controls in Internet Appendix Table IA.17, and our results remain robust.

Second, we analyze equity returns around key gold clause events to verify that the market reacted in accordance with our hypothesis. As detailed below, we find that gold-exposed firms experienced positive abnormal returns when the abrogation was passed (favorable news) and negative abnormal returns during the Supreme Court hearings (unfavorable news), with the pattern reversing upon the Court's favorable ruling. This evidence, which we discuss further in Section 3 of the manuscript, supports the view that markets perceived gold clause exposure as economically meaningful during the litigation period.

For some of this period Edwards et al. (2015) even has measures of market implied probability of the gold clause "mattering" based on the spread between treasuries with and without the clause. Higher frequency data, like would be available with secondary market equity prices, could help validate that effects line up with changes in the more cleanly identified movements in Edwards et al. (2015). Dividend payouts are also higher frequency than annual and not paid out at the same time of year for all firms. That should provide additional higher frequency evidence on whether dividend payouts really change at the times that are expected given the events of interest.

We agree that secondary market equity data can help validate our research design. While we cannot construct a daily "gold premium" series comparable to Edwards, Longstaff, and Marin (2015)—who exploit variation across Treasury bonds with and without gold clauses—we follow Kroszner (1999) and analyze equity returns around key gold clause events to test whether markets reacted consistently with our hypothesis. We also provide a detailed discussion of how our findings relate to Edwards, Longstaff, and Marin (2015) in our response to Minor

Comment 8 below.

Based on the event timeline in Edwards (2018), we identify three episodes: (i) the Joint Resolution of Congress abrogating the clause (May 26–June 5, 1933), which we interpret as favorable to gold-exposed firms; (ii) the Supreme Court hearings (January 8–February 17, 1935), during which the government's weak performance increased uncertainty, unfavorable to gold-exposed firms; and (iii) the Supreme Court ruling upholding the abrogation (February 18, 1935), favorable to gold-exposed firms. These events are described in detail in Section 3 of the manuscript.

Table R.I reports cumulative returns for a d-weighted portfolio (where firm j's weight is $w_j = d_j / \sum_k d_k$), the market portfolio, and abnormal returns using CAPM and Fama-French three-factor models.

Event Type	d-Weighted Portfolio	Market	Abnormal (CAPM)	Abnormal (FF3F)	Favorable to Issuer
Legislative	0.272	0.117	0.143	0.042	Yes
Pre-Judicial	-0.051	-0.044	-0.051	-0.020	No
Judicial	0.037	0.029	0.007	0.004	Yes

Table R.I
Cumulative returns and abnormal returns during key episodes

The return evidence is consistent with our interpretation. During the Legislative episode, the d-weighted portfolio returned 27.2% compared to 11.7% for the market, with positive abnormal returns under both CAPM (14.3%) and FF3F (4.2%). During the Pre-Judicial period, both the market (-4.4%) and the d-weighted portfolio (-5.1%) declined, with negative abnormal returns for gold-exposed firms. On the day of the favorable Supreme Court ruling, the d-weighted portfolio returned 3.7% versus 2.9% for the market, though the abnormal returns are modest (0.7% CAPM, 0.4% FF3F). These results echo Kroszner (1999), who also finds positive effects of the abrogation on gold-exposed firms' stock returns.

We incorporate this evidence in Section 3 of the revised manuscript to complement the historical narrative. We thank you for this suggestion, which provides independent validation that markets perceived gold clause exposure as economically significant during the litigation period.

Regarding the referee's suggestion to examine dividends at a higher frequency, we estimate our baseline dividend specification separately for each calendar quarter, using quarterly cash dividends as the dependent variable. Because dividends exhibit strong seasonal patterns and are sticky, this approach tests whether the timing of dividend increases aligns with the events of interest. The results are reported in Internet Appendix Table IA.14. Although the statistical evidence is somewhat weaker at this higher frequency, the evidence is strongest in the second quarter: the Q2 interaction coefficients for 1933 and 1934 are both 0.010, statistically significant at the 1% and 5% levels, respectively. This timing is consistent with the abrogation in June 1933 and the litigation that followed through 1934. We also examine the robustness of the annual dividend results to sample restrictions and alternative normalizations in Internet Appendix Table IA.15. As discussed in Section 5.3 of the revised manuscript, the dividend increase is concentrated among prior dividend payers and is robust across several alternative definitions of the dependent variable.

d. *<u>% of long-term debt with gold clauses:</u> I don't think there are, but if there are enough firms with bonds without gold clauses then it seems reasonable that the main specification should probably be the % of bonds (among those firms with bonds) outstanding with gold clauses. Given that the above is unlikely the most convincing evidence is likely the absence of any effects for preferred stock. Given that it would be helpful to see the results using just variation in the % of long-term debt (preferred stock + bonds) with gold clauses among those with either preferred stock or bonds. Just like with the primary design it would be helpful to see how within the sub-group those with or without any gold clauses (and pre period correlations with covariates) differ along other dimensions.*

Thank you for this suggestion, which became a key element of our revision. You are correct that the ideal variation would be across bonds with and without gold clauses. Since this approach is not feasible—nearly all bonds contained gold clauses, as explained above—we now define d as the share of gold clause bonds in total long-term debt, including preferred shares, following your recommendation. We discuss this methodological change in the data section (Section 4) and provide the explicit definition of d in equation (1). The largest component of long-term liabilities other than bonds is preferred shares; among firms with long-term liabilities, bonds and preferred shares account for approximately 96% of long-term liabilities, with the remainder consisting of items such as bank debt and bonds without gold clauses.

In our baseline tests, we continue to include firms that use neither bonds nor preferred shares for financing, assigning them $d = 0$. Following your suggestion, we also replicate the baseline results for the subsample of firms with bonds, preferred equity, or bank debt outstanding in 1930. These results are reported in Table 3 (Column 5) and discussed in Section 5.2 of the manuscript, with summary statistics provided in Internet Appendix Table IA.10. The results are consistent with our baseline findings: the interaction coefficients for 1933 and 1934 are -0.058 and -0.043, respectively, both statistically significant at the 1% level, compared to -0.060 and -0.041 in the baseline specification.

As you requested, Table 1 now reports pre-period characteristic differences between $d = 0$ and $d > 0$ firms.

Then historical discussions of why preferred don't include gold clauses and what drove decisions to issue one vs the other (ex. should we be concerned about differential exposure to the interest tax shield?). Any concerns addressed here are relevant for the original primary design, but alleviates many of the other concerns.

We discuss the historical origins of the gold clause and its differential adoption across security types in Section 3.1 of the manuscript. As we write there:

> "The gold clause emerged in the United States during the Civil War when gold-backed currency and fiat money (greenbacks) circulated simultaneously... By the

> time the US returned to the gold standard in 1879, gold clauses had become a standard feature in most corporate and utility bonds, as well as many farm and house mortgages... This selective use of the clause in long-term contracts can be seen among the large industrial firms studied in this paper. While the clause is present in the vast majority of corporate bonds (97% in our sample), it is entirely absent from preferred shares... The exact reason why gold clauses were included in nearly all corporate bonds, present in some mortgages, and absent from preferred shares remains unclear. Despite extensive investigation of contemporary sources and legal studies, we found no definitive explanation for this differential adoption pattern."

We conclude in the manuscript that "the widespread use of the gold clause in corporate bonds originated from the monetary disruptions of the Civil War. This historical accident would prove consequential 70 years later when the United States abandoned the gold standard in the midst of the Great Depression."

Regarding the factors driving the choice between preferred stocks and corporate bonds, we note that both were widely used in the 1930s. In our sample, preferred stocks and corporate bonds constituted roughly equal proportions of firms' capital structures, and firms frequently used both. While corporate bonds were tax advantaged relative to preferred shares, the US corporate tax rate was relatively modest at 13.75% at the time of the abrogation, limiting the potential for differential selection into bonds versus preferred shares based on tax considerations.

Despite these distinctions, preferred shares also represent a financing choice that commits firms to stable payments. While missing such payments is not legally equivalent to defaulting on a bond coupon, it would often entail higher financing costs and restrictions on common payouts (e.g., an "arrears overhang"). Thus, we believe that contrasting bonds with preferred shares remains useful, as both represent long-term fixed commitments by firms.

In fact, I would argue even ex-ante this source of variation is much more likely to have *plausible causal interpretation and the authors should seriously consider if an approach like this would be a more appropriate primary specification.*

We agree, and we have adopted this approach as our primary specification. As described in Section 4, our main measure d is now defined as gold clause bonds divided by long-term liabilities (bonds, preferred shares, and bank loans), rather than by total liabilities as in the previous draft. This definition, formalized in equation (1), is used throughout all analyses in the revised manuscript. We continue to include firms without long-term liabilities (assigning them $d = 0$) to preserve sample size, but the underlying measure follows your recommendation.

We also report results restricting the sample to firms with positive long-term liabilities in Table 3 (Column 5), as discussed above. Consistent with your intuition that "the most convincing evidence is likely the absence of any effects for preferred stock," we report placebo tests in Section 5.2 of the manuscript. As we write there:

> "We next conduct placebo tests to verify that the documented investment effect is specifically attributable to gold clause exposure. To do so, we re-estimate [our baseline specification] while replacing the numerator of $\tilde{d}$ with the book value of preferred shares. If the investment effect documented in our baseline specification results from gold exposure, we do not expect to find any effect when replacing the numerator of $\tilde{d}$ with preferred shares since the gold clause was not included in those securities. The results in column 6 support this conjecture: the interaction coefficients in 1933 and 1934 are positive and statistically insignificant."

The preferred shares coefficients (0.024 and 0.016) are economically small and statistically insignificant, in stark contrast to the large and highly significant negative coefficients for gold bonds (-0.060 and -0.041, both significant at the 1% level). This confirms that our baseline results are specific to gold clause exposure rather than long-term financing more broadly.

Major Comment 3.

Another substantial concern is the lack of clarity on what exactly "leverage risk" is, is it new, what it would predict, and if those differentially predictions are really demonstrated empirically. The authors mention debt overhang and also models of uncertainty, but those seem like they might generate completely opposing predictions. If there are costs of financial distress then it seems likely that firms would want to build-up cash, rather than engage in equity payouts, if they are concerned about future uncertainty (aka an increased probability of needing to make larger debt payments to avoid distress).

On the other hand, if it is about debt overhang then perhaps as the authors indicate firms could want to increase equity payouts. If debt overhang is the mechanism though, then there does already exist some empirical evidence in that space (ex. Bennett et al. 2018 and Wittry 2018 as recent examples) and it isn't clear in the way things are written now that leverage risk is empirically distinguishable from debt overhang in this setting. As the authors note in this setting conditional on the abrogation being overturned there could be a 69% increase in bond payments for exposed firms just after perhaps the most severe economic downturn in U.S. history. It seems entirely plausible this has a substantial effect on the probability of bankruptcy, and therefore debt overhang. Firms always have some "leverage risk" since the future value of the asset side of the balance sheet is always uncertain and that could generate debt overhang if firms are sufficiently levered. What then are the differential predictions when there is "leverage risk" coming from uncertainty in the future liability and how do the authors show evidence of that? If there aren't differential predictions, then it might be more appropriate to frame this paper as additional evidence on the effects of debt overhang on corporate investment and then compare/contrast much more with the existing literature/evidence in that space.

Thank you for this comment. We agree, based on feedback from the other reviewers as well, that the term "leverage risk" was confusing. We have removed it throughout the paper, including from the title, which is now "Debt Overhang During Policy Uncertainty: The Case of Gold Clauses in the 1930s."

As you correctly note, firms always face some degree of leverage risk due to uncertainty in the future value of assets, which can generate debt overhang if firms are sufficiently levered. Our setting differs in that the source of uncertainty arises on the liability side of the balance sheet—specifically, from the potential revaluation of gold clause obligations. We view the predictions from gold clause exposure as a direct application of debt overhang. To clarify this point, we added Section 5.1, which frames our analysis within the dynamic debt overhang model of Hennessy (2004). As we write in the manuscript:

> "For firms with long-term debt, Hennessy (2004) shows that capital structure creates a wedge $r_{j,t}$ between a firm's *average* Q and its *marginal* q... The gold clause litigation provides a natural setting to apply this framework. A ruling against the abrogation would have raised the claim of debtholders on the firm's assets and increased the probability of default. Accordingly, the legal uncertainty about the abrogation must have increased the expected recovery value $R_{j,t}$ and thus the debt overhang wedge $r_{j,t}$."

We interpret 1933 and 1934 as years when this wedge as a function of a firm's gold clause exposure.

We have also expanded our discussion of the broader debt overhang literature in Section 2, explicitly positioning our contribution relative to existing empirical work. As we now write in the introduction: "Our results complement the empirical literature on debt overhang that exploits plausibly exogenous variation in leverage to study firm decisions (e.g., Giroud, Mueller, Stomper, and Westerkamp (2011), Wittry (2020), Bennett, Ham, Milbourn, and Wang (2025)). With respect to this literature, our empirical setting has a unique feature: the abrogation created uncertainty about future leverage increases that ultimately did not materialize." This unique feature—that the liability increase never materialized—connects our work to the literature on firm-level uncertainty, which we now discuss more explicitly following R2's suggestions.

Regarding your concern about opposing predictions from debt overhang versus precautionary motives: we find that firms with higher gold exposure raised equity payouts rather than accumulating cash and reducing leverage. This pattern is consistent with Hennessy (2004)'s prediction that shareholders of distressed firms optimally extract value through dividends, and harder to reconcile with precautionary behavior that would predict lower payouts and cash hoarding. Together, this evidence suggests that debt overhang is the more likely operative mechanism.

Minor Comment 1.

It would be helpful if the authors explored the equity payout results in a bit more detail. I would encourage the authors to try a number of different definitions including the % change in payouts, dividend yield, % of profits paid out as dividends, and break out changes coming from cash dividends vs. share repurchases and intensive vs. extensive margins (not paying out at all vs. paying out only somewhat).

Using the extended dataset, we now report year-by-year results separately for total equity payout, cash dividends, and net repurchases (Table 4). The findings indicate that the effects are primarily driven by cash dividends rather than repurchases.

In Panel A of Internet Appendix Table IA.15, we report cash dividend results conditioning on whether firms paid positive or zero dividends in the prior year (Columns 1–2), and in 1932, just before the leverage uncertainty period (Columns 3–4). The results indicate that the dividend increase is concentrated among firms that were already paying dividends: columns 1 and 3 show strong and statistically significant effects for prior payers, while columns 2 and 4 show no significant effect for non-payers. This pattern is consistent with the well-documented stickiness of dividend policy.

We also followed your suggestion and experimented with alternative dividend measures (dividend growth, dividends relative to book equity, market equity, and net income) in Panel B of Internet Appendix Table IA.15. The results are generally consistent across these specifications, with positive but not always statistically significant point estimates. In particular, because many firms report negative net income, the sample in which dividends represent a positive share of profits is relatively small.

Minor Comment 2.

It would be useful to see the coefficient on d in Table 4 column 1 of Panels A and B.

This coefficient is now reported in all specifications.

Minor Comment 3.

What are the number of clusters in these regressions?

With the extended sample period (1926–1940), we now cluster standard errors by both firm and year in all specifications—whether using year indicators or grouped period indicators (Before and After). This yields 543 firm clusters, corresponding to the number of unique firms in the sample, and 15 year clusters. We have added this information in the footnotes of the relevant tables.

Minor Comment 4.

What do the results look like using just the d>0 indicator variable that matches more closely to what is done in Figure 1?

This result is now reported in Column 1 of Internet Appendix Table IA.16.

Minor Comment 5.

What are the correlations (and statistical significance) between d and co-variates in 1931–1932 period?

We report the correlations between d and other firm characteristics in the Internet Appendix Table IA.8. Because our new sample starts in 1926 rather than 1931, we report the correlations both for the 1926–1932 period as well as 1932 (hence, right before the leverage uncertainty period). As suggested by the significant differences between the $d = 0$ and $d > 0$ samples in Table 1, all correlations are statistically significant except only one for the 1932 sample.

Minor Comment 6.

How do the authors reconcile Figure 1 and results from the rest of the paper? Figure 1 shows as the authors note that "[w]e find that leverage risk led public firms to delay investment in fixed capital for two years." That would suggest that not only does the period in 1933-1934 have a lower net investment rate than 1931-1932, but that 1935-1936 has a higher net investment rate than 1931-1932 and enough higher that the cumulative effects are 0 over this whole period. In other words, no investment is lost, just delayed. By contrast, Table 2 and 4 both suggest that 1935-1936 net investment is going to be basically identical to that in 1931-1932. That would suggest there is a cumulative effect as 1935-1936 net investment does not offset the decline in 1933-1934. It *would be helpful for the authors to explicitly test 1935-1936 relative to 1931-1932 and provide some discussion for reader about which findings (the table or figure) seem most relevant and what the delay vs loss means in terms of the theories being tested (ex. Mello and Parsons 1992 would seem to suggest a delay). One natural prediction would seem to be that the reduction in uncertainty in 1935-1936 should lead to the same net investment difference as would occur in 1931-1932, plus some additional positive net investment for those treated firms. The current evidence in the main paper/tables doesn't seem to support that conclusion.*

Thank you for this comment. You are correct that our earlier evidence was more consistent with a return to usual investment levels rather than a delay. Our revised empirical design, where we report the coefficient on d interacted with year indicators, makes this clearer. Table 3 and Figure 3 of the revised manuscript report net investment results from 1926 to 1940, with d interacted with year indicators. Using 1932 as the omitted category, we find that higher d predicts lower investment only in 1933 and 1934. Column 2 of Table 3 shows some evidence of a statistically significant recovery for exposed firms in 1935. After 1935, investment differentials across d converge back to pre-1933 levels. We now briefly note this finding in Section 5.2 (third paragraph).

Minor Comment 7.

If the gold clauses hurt firm value, how come we don't see that in Tobin's Q? It would be helpful to show explicitly that the equity value of these firms are hurt by the risk during this period or if the authors find it isn't provide a plausible explanation for why.

Thank you for this observation. As clarified in Section 5.1, our conceptual framework emphasizes that debt overhang creates a wedge between marginal and average Q. Accordingly, investment should be lower for firms with more severe debt overhang, even after controlling for average Q—which is precisely what we find in the data.

That said, you are right that we might also expect a larger decline in average Q in 1933 and 1934 for firms with higher gold clause exposure, but we do not find this. In Table R.II, we regress Q on d interacted with four indicators—1926–1932 (Before), 1933, 1934, and 1935–1940 (After)—with Before as the omitted category. If anything, high-d firms experienced a slightly smaller decline in average Q in 1933. When we add controls interacted with year, as in Column 2 of Table 6, the coefficient for 1933 becomes negative but remains relatively small.

	Without controls	With controls
1933 $\times d$	0.18**	-0.06*
	(0.06)	(0.03)
1934 $\times d$	0.01	-0.02
	(0.08)	(0.04)
After $\times d$	0.02	0.03
	(0.10)	0.05

Table R.II
Average Q

The return evidence reported in our previous response suggests that the equity value of the gold-exposed firms was indeed affected by the uncertainty of leverage generated by the abrogation. When we narrow down our analysis of equity value to important events related to the enforceability of the gold clause—such as when Congress passed the Joint Resolution in June 1933 and when the Supreme Court rendered its decision—we find that the returns (raw and abnormal) on gold-exposed firms were significantly larger than those of the overall market.

Identifying why average Q was unaffected in 1933 and 1934 is admittedly difficult. However, if we interpret average Q as a measure of investment opportunities, the absence of a differential effect on Q suggests that the abrogation did not differentially impact the investment opportunities of exposed firms. Rather, the investment decline we document reflects the reduced incentive of exposed firms to invest due to debt overhang: even in the presence of similar investment opportunities, equityholders of exposed firms had weaker incentives to fund new projects because a larger share of the returns would accrue to bondholders in the event of default.

Minor Comment 8.

How should we reconcile the findings in this paper with those in Figure 4 of Edwards et al. (2015)? They find no effect of gold clauses until 1934, while this paper finds evidence in 1933. They find no reduction in the risks of the gold clause in 1935, while this paper suggests there are. It would be good to include a detailed discussion of these divergent assumptions/findings and why the authors don't *believe they are problematic. At its surface high frequency data on prices for otherwise virtually identical bonds with and without gold clauses trading on the same days seems like a fairly clean approach to examine the pricing effects of gold clauses, so it seems important to reconcile with those findings.*

Thank you for raising this point. To address it, we carefully reread Edwards, Longstaff, and Marin (2015) to better understand the source of the difference in our results. We have identified several possibilities. Before expanding on them, we begin with a brief summary of Edwards, Longstaff, and Marin (2015)'s results.

In their paper, Edwards, Longstaff, and Marin (2015) study how the abrogation of the gold clause has affected the Treasury's borrowing cost. To do so, they compare the yields of a portfolio of bonds that were issued before the abrogation of the gold clause (which included the gold clause) to the yield of bonds that were issued after the abrogation of the gold clause (which did not include the gold clause). Because the bonds in these two groups have different effective durations, the portfolios are dynamically rebalanced daily over their sample to match the effective duration. For convenience, we report the time series of the yield spread reported in Figure 4 of their article in Figure R.II below. In this figure, the y-axis is the yield difference between bonds without a gold clause and bonds with a gold clause,

$$y = \text{portfolio yield}_{\text{Without Gold}} - \text{portfolio yield}_{\text{With Gold}}.$$

A positive value of y indicates a lower yield on gold bonds, which in turn corresponds to a price premium on bonds with a gold clause.

As noted in your comment (as well as by Edwards, Longstaff, and Marin (2015) themselves), a striking feature of this plot is the absence of any reduction in the gold premium following the Supreme Court's February 1935 decision to uphold the abrogation of the gold clause. In fact, the spread appears to widen in the aftermath of the Court's decision, a puzzling observation for which we propose two potential explanations.

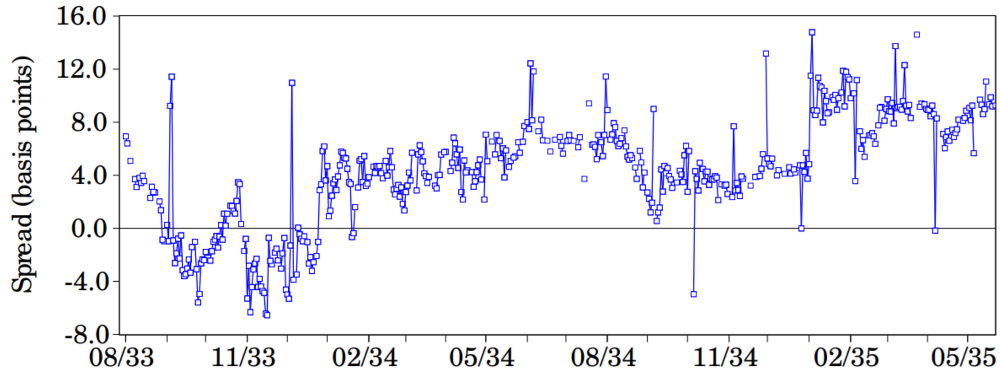

Figure R.II. Yield spread between gold and non-gold denominated bonds

The first explanation requires additional historical context regarding the Supreme Court's ruling on the abrogation's constitutionality. The Court's decision upholding the abrogation applied to both private and public contracts, but the justices employed distinct legal reasoning for each. For private contracts, the Court ruled 5-4 that the abrogation was constitutional. However, for public contracts, the Court ruled 8-1 that the abrogation was *un*constitutional. Ironically, despite ruling the abrogation unconstitutional, the Court determined that the plaintiffs had not suffered recoverable damages and therefore awarded no compensation.

It seems plausible that US Treasury bondholders interpreted the decision as leaving the door open for future favorable rulings should public bond holders succeed in proving damages. Supporting this interpretation, Edwards, Longstaff, and Marin (2015) notes that in the years following the Court's decision, additional lawsuits challenging the constitutionality of the abrogation were filed, though the Supreme Court declined to hear these cases. For convenience, we reproduce the relevant passage from Edwards, Longstaff, and Marin (2015) below:

"Curiously, the premium for gold clause bonds persisted even after the Supreme Court decision. One possible interpretation of this result is that investors continued to believe that a return to the gold standard might occur once macroeconomic fundamentals improved. Another possible interpretation is that the market believed there was some probability that the Supreme Court would reverse itself and grant some compensation to holders of gold clause securities. Indeed, during the years to come, a number of lawsuits regarding the abrogation made it through the court system. The Supreme Court, however, refused to hear any of these cases."

To provide additional detail on the legal differences between private and public contract cases, we have added a footnote to the "Historical background" section. This footnote reads "The Court heard four cases: two concerning private contracts and two concerning public contracts. For private contracts, the Court ruled 5-4 that the abrogation was constitutional. For public contracts, the Court ruled 8-1 that the abrogation was *un*constitutional. However, despite finding the abrogation of gold clauses in public contracts unconstitutional, the Court determined that the plaintiffs had not suffered recoverable damages and thus awarded no compensation." We abstain from providing a detailed discussion of the subtlety of the Supreme Court's ruling in that section to keep the focus on the background information most relevant to our aim.

Before expanding on the second set of factors that might explain the difference between our results and those of Edwards, Longstaff, and Marin (2015), we should note that the yield comparison exercise performed in their paper is deceptively challenging. Extracting the market value of the gold clause by comparing the yield of two dynamically managed portfolios of Treasury bonds is difficult since US Treasury bonds in the 1930s were highly bespoke securities (see Lehner, Payne, Shurtleff, and Szőke (2025) for a comprehensive treatment of the measurement challenges involved in constructing historical Treasury yield spreads). We expand below on the idiosyncratic attributes of US Treasuries during this period, emphasizing how these features may create yield differentials between the two bond types that are unrelated to the gold clause.

First, as Edwards, Longstaff, and Marin (2015) note in their paper, 9 out of 10 bonds in their sample were callable. Moreover, as they also indicate, most call options for the gold-denominated bonds were deep in the money. In contrast, the non-gold denominated bonds had just been issued and their call options were likely out of the money. This implies that changes in yields over their sample period might reflect changes in call option values in addition to changes in the value of the gold clause.

Second, the US Treasury was issuing bonds with various tax statuses: some bonds were fully taxable, some were partially taxable, and some were nontaxable. This raises the possibility that the difference between gold and non-gold denominated bonds reflects differences in tax treatment rather than gold clause valuation. Edwards, Longstaff, and Marin (2015) do not address this issue in the most recent version of the paper I have access to.

Third, the US Treasury was also issuing so-called "flower bonds." These bonds were redeemable at par value upon the holder's death and provided estate tax benefits. This raises the possibility that differences between gold and non-gold denominated bonds reflect the varying prevalence of flower bonds in the sample considered by Edwards, Longstaff, and Marin (2015). This issue is also left unaddressed in their analysis.

To summarize, we believe that differences in the legal context of the abrogation between public and private gold clauses may explain part of the discrepancy between our results and those of Edwards, Longstaff, and Marin (2015). Moreover, the yield comparison exercise performed in their paper is potentially affected by measurement issues. These issues remain unaddressed in the most recent version of Edwards, Longstaff, and Marin (2015) available to us, which complicates interpretation of the evidence presented above.

Thank you again for your constructive and insightful comments. The new analyses, guided by your suggestions, have significantly strengthened the empirical evidence and enhanced our understanding of the mechanism, providing a more comprehensive perspective on this period.

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